Dec. 4

Student Hours

This week as usual 5:45 pm W, R MLC

Friday after class: no student hours

Next week

T, W Tues: 4:30 Wed: 5:45 MLC

Final Exam Thurs. Dec. 12 10:30 am - 12:30 pm SPARK 333

Properties of 1-Dim. Heat Eq'n $u_t - u_{xx} = 0$.

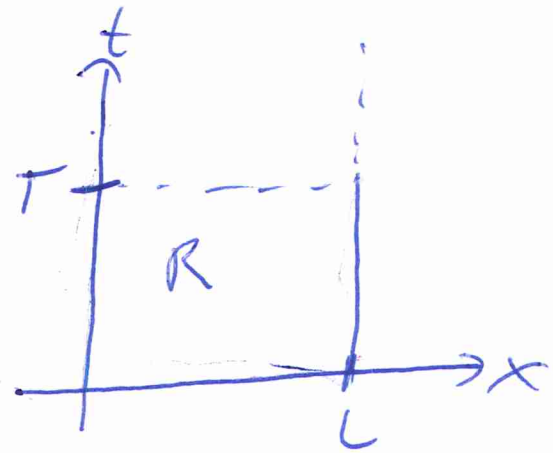
① Max-Min Thm

Let T be any number >0.

Suppose that $u \in C^0(\bar{R})$

+ $u_t - u_{xx} = 0 \quad 0<x<L,\ 0<t\le T$

Then u attains its max + min values on the lower base, $t=0$, or on the vertical sides, $x=0, x=L$ of R.



$R = \{(x,y) : 0<x<L,\ 0<t<T\}$

Domain of heat eq'n is $\sqcup$ $(t>0)$

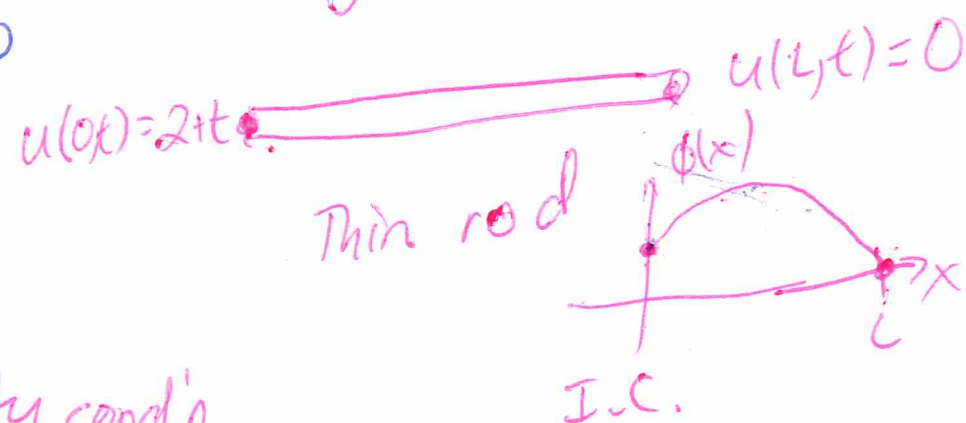
$\Rightarrow$ gives us uniqueness + cont dep. on data.

② Smoothness of sol'n.

Consider

$$(*)\begin{cases} u_t - u_{xx} = 0 & 0 \le x < L,\ t>0 \\ u(x,0) = \phi(x) & 0 \le x \le L \\ u(0,t) = 0 \quad u(L,t) = 0 & t \ge 0 \\ \phi \in C^0([0,L]),\ \underbrace{\phi(0) = \phi(L) = 0}_{\text{compatibility cond'n}} \end{cases}$$

Physical Problem



The sol'n to (*) is C^∞ for $t>0$.

What problems are well-posed?

(*) is well-posed (Dirichlet B.C.)

Heat eq'n on ∞-long domain is also well-posed. Sep. of variables

on semi-∞-long domain " " " "

$0<x<\infty$, If $u(0,t)=0$

Then extend $\phi(x)$ <u>oddly</u> + use Sep. of variables

If $u_x(0,t)=0$ extend I.C. f'n evenly

+ use S. of V.

What about Neumann BC? ∞-many sol'ns? Let's see.

HW#15 problem 4

$u_t - u_{xx} = 0 \quad 0 < x < L, \; 0 < t$

$u(x,0) = \phi(x) \quad 0 \le x \le L \qquad \phi(x) \in C^0([0,L])$

$u_x(0,t) = 0 \quad u_x(L,t) = 0 \qquad 0 \le t$

compatibility? No worries. $\phi(x) \in C^0$.

Solve by S. of V. assume a sol'n of the form $u(x,t) = X(x)T(t)$

① Subst. into eq'n:

$u_x = X'(x)T(t) \qquad u_{xx} = X''(x)T(t) \qquad u_t = X(x)T'(t)$

$$\left[X(x)T'(t) - X''(x)T(t) = 0\right] \frac{1}{X(x)T(t)}$$

② separate

$$\frac{X(x)T'(t)}{X(x)T(t)} - \frac{X''(x)T(t)}{X(x)T(t)} = 0$$

$$\frac{T'}{T} - \frac{X''}{X} = 0$$

$$\frac{T'}{T} = \frac{X''}{X} = \mu$$

which order to solve? Look at BC / IC

$u(x,0) = \phi(x)$ $\quad$ $u(x,0) = X(x)T(0) = \phi(x)$

no information

$u_x(0,t) = X'(0)T(t) = 0$

either $X'(0) = 0$ or $T(t) = 0 \Rightarrow u = 0$

trivial sol'n ↓

$\Rightarrow X'(0) = 0$

$u_x(L,t) = X'(L)T(t) = 0 \Rightarrow X'(L) = 0$

We have 2 conditions for $X(x)$, so work on this eq'n first.

$$X'' - \mu X = 0$$

$\mu > 0$ $\quad X(x) = c_1 e^{\mu x} + c_2 e^{-\mu x}$ ⑥

$X'(x) = c_1 \mu e^{\mu x} + c_2(-\mu) e^{-\mu x}$

Using BC we get $c_1 = c_2 = 0$ Trivial Sol'n ☹

$\mu = 0$ $\quad X(x) = c_1 + c_2 x$

$X'(x) = c_2$

using BC get $c_2 = 0$ & c_1 can be anything. However

$\Rightarrow u(x,t) = c_1 T(t)$ & I.C. $u(x,0) = c_1 T(0) \neq \phi(x)$ for all $\phi(x)$.

Trouble ☹

$\mu < 0$ say $\mu = -\lambda^2$.

Then $X(x) = c_1 \sin(\lambda x) + c_2 \cos(\lambda x)$

$X'(x) = c_1 \lambda \cos(\lambda x) - c_2 \lambda \sin(\lambda x)$

$X'(0) = c_1 \lambda = 0 \Rightarrow c_1 = 0$

$X'(L) = -c_2 \lambda \sin(\lambda L) = 0$

$\Rightarrow c_2 = 0 \ (\because)$ or $\sin(\lambda L) = 0$

$\Rightarrow \lambda L = 0, \pm\pi, \pm 2\pi$

so $\lambda = \frac{n\pi}{L}$ $n = 0, \pm 1, \pm 2, \ldots$

$X_n(x) = \cos\left(\frac{n\pi}{L} x\right)$ $\mu = -\left(\frac{n\pi}{L}\right)^2$

solve for $T(t)$

$T' - \mu T = 0$ or $T' + \left(\frac{n\pi}{L}\right)^2 T = 0$

$T_n(t) = e^{-\frac{n^2\pi^2}{L^2} t}$

Put all together

$u_n(x,t) = e^{-\frac{n^2\pi^2}{L^2} t} \cos\left(\frac{n\pi}{L} x\right)$

General sol'n By principle of superposition (8)

$$u(x,t) = \sum_{n=0}^{\infty} C_n e^{-\frac{n^2\pi^2}{L^2}t} \cos\left(\frac{n\pi}{L}x\right)$$

I.C. $u(x,0) = \sum_{n=0}^{\infty} C_n \cos\left(\frac{n\pi}{L}x\right) = \phi(x)$

C_n's are Fourier cosine series of $\phi(x)$ coeff.